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Method for Operating a Measuring Device

Abstract

A computer-implemented method for operating a measuring device, in particular a wall diagnostic device, includes (i) receiving radar data of a radar sensor unit of the measuring device, (ii) performing wall diagnostics by performing an analysis of the radar data and generating diagnostic results using a diagnostic module of the measuring device, (iii) performing an object recognition of an object disposed in the wall using the diagnostic module based on the radar data and taking into account the wall type classification results of the wall type, and (iv) providing the diagnostic results of the diagnostic module to a display unit of the measuring device for displaying the diagnostic results to a user of the measuring device.

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Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 237.3, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

[0002] The present disclosure relates to a method for operating a measuring device, in particular a wall diagnostic device.

BACKGROUND

[0003] Diagnostic devices for diagnosing walls and detecting objects formed in the walls are known in the prior art.

[0004] It is an object of the present disclosure to provide an improved method of operating a measuring device, in particular a wall diagnostic device.

[0005] The task is solved by the process set forth below. Advantageous embodiments are subject-matter also set forth below.

SUMMARY

[0006] According to one aspect, a computer-implemented method of operating a measuring device, in particular a wall diagnostic device is provided, wherein the method comprises: [0007] receiving radar data from a radar sensor unit of the measuring device, wherein the radar data depicts a wall to be diagnosed; [0008] performing a wall diagnostics by performing an analysis of the radar data and generating diagnostic results via a diagnostic module of the measuring device, wherein the wall diagnostic comprises: [0009] performing a wall type classification and providing wall type classification results of a wall type of the wall using the diagnostic module based on the radar data; and [0010] performing an object recognition of an object disposed in the wall using the diagnostic module based on the radar data and taking into account the wall type classification results of the wall type, wherein the object recognition comprises an object detection and an object classification; and [0011] providing the diagnostic results of the diagnostic module to a display unit of the measuring device for displaying the diagnostic results to a user of the measuring device, wherein the diagnostic results comprise at least one object position of the object in the wall and an object type of the object and/or the wall type of the wall.

[0012] The technical advantage can thereby be achieved that an improved method for operating a measuring device, in particular a wall diagnostic device, can be provided. To this end, based on radar data from a radar sensor unit of the measuring device that depicts a wall to be examined, a wall diagnostics is performed by performing a diagnostic module of the measuring device.

[0013] The wall diagnostics comprises performing a wall type classification, whereby the diagnostic module determines a wall type of the wall to be examined, and performing an object recognition, whereby the diagnostic module performs object detection and object classification of an object disposed in the wall based on the radar data and taking into account the previously determined wall type. Further, diagnostic results are provided for the wall diagnostics performed by the diagnostic module, wherein the diagnostic results comprise at least the wall type of the wall to be examined and/or an object position or object type of the object positioned in the wall.

[0014] By automatically determining the wall type of the wall to be examined, a more precise wall diagnostics can be provided. Further, it may be avoided that a wall type must first be determined

manually by the user prior to performing the wall diagnostics. In this context, it may further be avoided that the user sets the wrong wall type, which would adversely affect the diagnostic results. By incorporating the wall type automatically determined by the diagnostic module into the object recognition, a more precise object recognition can be provided.

[0015] Different wall types affect the radar data received from the measuring device, so that effects of the wall type, for example in a subsurface correction, can be taken into account or corrected for the radar data in the object, in order to be able to perform the object recognition independent of the present wall type. This may increase the quality of the wall diagnostics performed.

[0016] According to one embodiment, the received radar data comprises a plurality of radar signals of different frequencies reflected by the wall, and wherein the received radar data is summarized into a plurality of windows.

[0017] The technical advantage can thereby be achieved that, by evaluating the received radar data in a plurality of spatially larger windows, wall diagnostics may be performed based on the radar data for a larger spatial area. This accelerates performance of the wall diagnostics.

[0018] According to one embodiment, performing the wall type classification comprises: [0019] performing partial classifications on individual windows of the radar data and providing partial classification results of the wall type; and [0020] merging a plurality of partial classification results to an overall classification result and determining the wall type based on the overall classification result.

[0021] The technical advantage can thereby be achieved that a more precise wall diagnostics can be provided. For this purpose, a partial classification is performed based on the individual windows of the radar data, and partial classification results of the wall type are provided for each window. Subsequently, multiple contiguous partial classification results are merged to an overall classification result of the wall type, and the final wall type is determined based on the overall classification result.

[0022] By performing a stand-alone partial wall type classification for each window, precise partial wall type classification results can be generated. Particularly when moving the measuring device relative to the wall, during which the radar data is received or processed in the form of windows over a period of time, meaning that each depicts different positions of the wall, the partial classification for different positions of the measuring device relative to the wall and associated different areas of the wall can thus be used to independently determine the respective wall type.

[0023] This allows, on the one hand, a precise determination of the wall type and, on the other hand, detection of a change of the wall type in different areas of the wall to be examined. By merging the data, the respective wall type of the wall to be examined can be determined as the overall classification result. For example, when the wall type changes in different areas of the wall, the merger may comprise a corresponding weighting to account for any wall type changes that may be present or possible erroneous diagnostic results.

[0024] According to one embodiment, the merger is effected by performing a merger algorithm, and/or wherein the merger takes into account the partial classification results with a weighting with respect to the results of the object recognition.

[0025] The technical advantage can thereby be achieved that the merger algorithm or the weighted consideration of the partial classification results in the merger can take into account the quality of the individual partial classification results. The quality of the overall classification result or the quality of the wall type of the wall ultimately determined can thereby be increased. Partial classification results that deviate from the majority of the partial classification results may be taken into account with a lower weighting, respectively.

[0026] According to one embodiment, performing the wall type classification comprises: [0027] storing the wall type classification results of the wall type; [0028] comparing currently determined wall type classification results of the wall type with stored wall type classification results of the wall type; and [0029] providing the current wall type classification results of the wall type if the

current wall type classification results of the wall type are different from the stored wall type classification results of the wall type for a predetermined number of windows of the radar data. [0030] The technical advantage can thereby be achieved that the wall type can be determined precisely. To this end, wall type classification results generated at earlier times are stored and wall type classification results generated at current times are compared with the stored wall type classification results.

[0031] If there is a deviation between the current wall type classification results, which correspond to the most recently determined wall type classification results, and the stored wall type classification results, which correspond to the respectively older wall type classification results, then the current wall type classification result which deviates from the stored wall type classification results, is only considered the actual wall type present if the current wall type classification result which deviates from the stored wall type classification results is confirmed by further measurements, i.e., wall type classification results determined later.

[0032] This may result in wall type classification results based on, for example, erroneous radar data or a faulty wall diagnostics which deviate from the previously determined wall type classification results due to this deviation being detected as erroneous and/or non-representative wall type classification results, and thus being disregarded. This may avoid erroneous wall diagnostics results.

[0033] Further, by constantly comparing the currently generated wall type classification results with the previously determined wall type classification results, any change in the present wall type of the wall to be examined can be determined and considered. This is particularly interesting in cases where large areas of the wall are examined by moving the measuring device along the wall. Because the change in wall type is taken into account in the wall diagnostics, a more precise wall diagnostics and in particular a more precise object recognition of the objects disposed in the wall can be achieved.

[0034] According to one embodiment, the radar data is received for a plurality of positions of the measuring device relative to the wall, wherein the method further comprises:

[0035] receiving movement data of a movement detection unit of the measuring device, wherein the movement data represents a movement of the measuring device relative to the wall between the plurality of positions of the measuring device relative to the wall.

[0036] The technical advantage can thereby be achieved that the relative movement of the measuring device relative to the wall can be detected by the movement data of the movement detection unit. By detecting the movement of the measuring device relative to the wall, an exact positioning of the measuring device relative to the wall can be determined. The positioning of the measuring device relative to the wall allows for precise object detection of objects over an extended area of the wall to be examined.

[0037] According to one embodiment, the method furthermore comprises: [0038] checking, based on the movement data of the movement detection unit, whether a movement of the measuring device relative to the wall by a predefined distance has been detected; [0039] if no movement has been detected, storing the radar data and performing the check once again; and [0040] If movement by the predefined distance has been detected, performing the wall diagnostics based on the radar data using the diagnostic module.

[0041] The technical advantage can thereby be achieved that a precise wall diagnostics is enabled over an extended area of the wall to be examined. Here, a check is first carried out to determine whether a movement of the measuring device relative to the wall by a predefined distance has been detected based on the movement data of the movement detection unit. If such movement has not been detected, the radar data is initially stored and the check is performed once again.

[0042] If such movement has been detected, the wall diagnostics are performed based on the radar data using the diagnostic module. This allows the wall to be scanned in a step size defined by the predetermined distance. This allows for reliable wall diagnostics over an extended area of the wall.

[0043] According to one embodiment, the wall diagnostics further comprises: summarizing sub-objects detected and classified to a contiguous object based on radar data received at different positions of the measuring device relative to the wall, if there is a predefined distance dependency between the sub-objects.

[0044] In this way, the technical advantage can be achieved that by summarizing the sub-objects detected and classified relative to the wall in the various positions of the measuring device based on the radar data into a contiguous object, a precise wall diagnostics is possible, including a precise object recognition. While moving the measuring device relative to the wall, the wall diagnostics are performed according to the embodiments described above.

[0045] For the movement segments defined according to the predefined distances, the partial results, i.e. partial objects of the object to be detected within the wall, are determined by the diagnostic module during the wall diagnostics. The classified partial objects here describe portions of an extended object that is depicted by movement of the measuring device relative to the wall in portions of the movement. By summarizing the correspondingly classified partial objects into an overall object, the extended object can be described and detected in detail. This allows for precise wall diagnostics and detection of extended objects.

[0046] According to one embodiment, the wall diagnostics further comprises: performing an object depth determination using the diagnostic module and determining an object depth of the object in the wall, wherein the object depth is defined as a distance of the object to a surface of the wall.

[0047] The technical advantage can thereby be achieved that, based on the radar data of the radar sensor unit, an object depth of the object within the wall can be determined by the measuring device in addition to the object recognition. The object depth describes a distance of the object to a surface of the wall. The correspondingly configured diagnostic module may determine the object depth based solely on the radar data of the radar sensor unit.

[0048] According to one embodiment, the analysis of the radar data further comprises: performing an object extension determination and determining an object extension of the object along a predefined direction via the diagnostics module.

[0049] The technical advantage can thereby be achieved that an object extension of the object detected in the wall is further enabled based on the radar data of the radar sensor unit. The object extension describes an extension of the object at least with respect to one spatial direction, preferably with respect to two spatial directions, especially preferably with respect to three spatial directions. This allows for a one-dimensional, preferably two-dimensional, more preferably three-dimensional extension information for the object located in the wall.

[0050] According to one embodiment, the wall diagnostics further comprises: [0051] performing a pre-processing of radar data of the radar sensor unit and providing input data to a wall type classification module of the diagnostic module using a first pre-processing module; [0052] performing the wall type classification based on the input data provided by the first pre-processing module and providing wall type information using a wall type classification module of the diagnostic module; [0053] performing a pre-processing of the radar data and providing input data to an object recognition module of the diagnostic module in consideration of the wall type information using a second pre-processing module of the diagnostic module; [0054] performing an object recognition based on the input data provided by the second pre-processing module and providing diagnostic results using the object recognition module.

[0055] The technical advantage can thereby be achieved that precise wall diagnostics are enabled by the correspondingly configured diagnostic module using the radar data of the radar sensor unit. By pre-processing the radar data of the radar sensor unit, the radar data can be brought into the form required for the wall diagnostics. Using a correspondingly configured wall type classification module, a corresponding wall type classification can be performed on the pre-processed radar data and the respective wall type of the wall can be determined.

[0056] An object recognition of an object disposed in a wall may be performed by an object recognition module based on the pre-processed radar data and taking into account the wall type provided by the wall type classification module. The object recognition module is configured to perform an object detection and object classification of the object disposed in the wall based on the pre-processed radar data and taking into account the provided wall type. The presented architecture of the diagnostic module allows for the most precise and reliable wall diagnostics for walls to be examined.

[0057] According to one embodiment, the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor for providing additional sensor data, wherein the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.

[0058] The technical advantage can thereby be achieved that, through further sensor data, additional sensors, each of which is configured to detect different physical variables, can provide additional information in addition to the radar data of the radar sensor unit, for incorporation into the wall diagnostics. This additional information, which is preferably complementary to the information of the radar data of the radar sensor unit, allows further precise specification of the wall diagnostics or the object recognition.

[0059] According to one embodiment, the diagnostic module comprises at least one correspondingly trained artificial intelligence configured to perform an object recognition and/or wall classification and/or object depth determination based on the radar data and/or the additional sensor data.

[0060] The technical advantage can thereby be achieved that, because the diagnostic module is formed as a correspondingly trained artificial intelligence, that is trained based on the radar data, or, if applicable, taking into account the information of the additional sensors, to perform an object recognition and/or a wall classification and/or an object depth determination and/or an object extension determination, a reliable and powerful diagnostic module can be provided. By using the artificial intelligence technique, precise wall diagnostics can be provided.

[0061] Provided according to one aspect is a computing unit configured to perform the method of operating a measuring device according to one of the preceding embodiments.

[0062] Provided according to one aspect is a computer program product comprising instructions that, when the program is executed by a data processing unit, cause the data processing unit to perform the method for operating a measuring device according to one embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0063] Embodiments of the disclosure are described with reference to the following figures. The figures show:

[0064] FIG. 1 a schematic illustration of a measuring device according to one embodiment;

[0065] FIG. 2 a further schematic illustration of the measuring device according to a further embodiment;

[0066] FIG. 3 a further schematic illustration of the measuring device according to a further embodiment;

[0067] FIG. 4 a schematic illustration of a measurement of the measuring device according to one embodiment,

[0068] FIG. 5 a further schematic illustration of the measuring device according to a further embodiment;

[0069] FIG. 6 a further schematic illustration of a measurement of the measuring device according

to one embodiment,

[0070] FIG. 7 a flowchart of a method for operating a measuring device according to one embodiment of the disclosure,

[0071] FIG. 8 a further flowchart of the method for operating a measuring device according to a further embodiment, and

[0072] FIG. 9 a schematic representation of a computer program product.

DETAILED DESCRIPTION

[0073] FIG. 1 shows a schematic representation of a measuring device **100** according to one embodiment.

[0074] The present disclosure relates to a measuring device, in particular to a wall diagnostic device for examining walls **105** to be processed. Wall diagnostic devices are known in the prior art that are used to detect objects disposed in walls. Such devices allow a user to examine walls to be processed to search for objects disposed in the walls in order to be able to perform planned work, for example drilling in walls, based on this such that damage to the objects disposed in the walls can be avoided.

[0075] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** to various operating modes.

[0076] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined and radar signals reflected by the wall **105** may be received.

[0077] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.

[0078] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform a corresponding diagnosis of the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** disposed within the wall **105**.

[0079] The wall diagnostics carried out by the diagnostic module **107** comprises at least performing an object recognition. The object recognition here comprises an object detection and an object classification of the object **113** disposed in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position here describes the positioning of the object disposed in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.

[0080] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** disposed in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.

[0081] The object **113** can be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.

[0082] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.

[0083] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.

[0084] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.

[0085] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105** and radar signals reflected from the wall **105** or the objects **113** disposed there are received via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.

[0086] For example, diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** disposed in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.

[0087] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.

[0088] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.

[0089] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than that given by the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0090] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.

[0091] According to its embodiment, the diagnostic module **107** is configured as a correspondingly

trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-described wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** disposed in the wall **105**. The object classification or determination the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.

[0092] The object classes may comprise: Metal/non-metal object, low voltage cable, single phase AC signal cable, multi phase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.

[0093] Further, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**. Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.

[0094] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.

[0095] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can in particular be used for object recognition of the objects **113** disposed in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.

[0096] In particular, for example, the material of the objects **113**, for example as a metallic or non-metallic material, can be improved and classified by using the additional sensor information.

[0097] FIG. 2 shows another schematic representation of the measuring device **100** according to a further embodiment.

[0098] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing of the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.

[0099] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall to be examined **105** and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.

[0100] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be included in the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0101] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137**

comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. In the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.

[0102] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.

[0103] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects on the radar data **103** can occur.

[0104] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **131** performs the object recognition of the object **113** disposed in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.

[0105] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.

[0106] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed

radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.

FIG. 3 shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0107] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, for example, comprising the wall type classification module **129** and/or the object recognition module **131** according to the embodiment in FIG. 2, and a post-processing module **163**.

[0108] In FIG. 3, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information of the additional sensors may also serve as input data for the wall diagnostics. Here, the different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual sub-analysis results to form the diagnostic results **109** of the wall diagnostics.

[0109] Alternatively or additionally, different sub-aspects of the wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0110] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moved relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0111] The different processing paths perform a stand-alone wall diagnostics in this respect, comprising at least determining the object position **115** and/or the object type **117** of the object **113** disposed in the wall **105**.

[0112] The summary module **165** may summarize the sub-results provided in the individual processing paths **102** of the stand-alone wall diagnostics of the different regions of the wall **105** into a contiguous diagnostic result **109**. The contiguous diagnostic result here describes the wall diagnostics of a contiguous spatial area that was covered during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.

[0113] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object disposed in the wall **113** can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0114] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may respectively be carried out. In the summary module **165**, the various sub-results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0115] The diagnostic module **107** can be divided into different artificial intelligences **125**, as already shown in the embodiment in FIG. 2. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0116] The respective modules may each be configured as stand-alone artificial intelligences **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known in the prior art.

[0117] FIG. 4 shows a schematic illustration of a measurement of the measuring device **100** according to one embodiment.

[0118] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Further, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate the corresponding sensor data may be sampled. Further, the radar data **103** or the additional sensor information **104** may be transformed to the respective required frequency range or time range. Methods known from the

prior art can be used for this purpose.

[0119] Further, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0120] Graph a) of FIG. **4** shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.

[0121] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement of the measuring device **100** ends.

[0122] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.

[0123] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.

[0124] A further temporal window **167** or spatial window **167** can be provided as soon as one or more scan points are available.

[0125] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. **3** as well. According to the embodiment of FIG. **2**, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.

[0126] As stated above, wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by rule-based algorithms. A combination of artificial intelligence and a rule-based algorithm is also possible within a processing path **102** in the form of a parallel connection or concatenation.

[0127] The diagnostic results **109** of the wall diagnostics may be implemented as numeric values, vectors, or matrices. Further, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be given for the object detection. The same may apply to the position and/or depth determination, for which corresponding probability values can also be given.

[0128] If, in addition to the radar data **103**, the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by rule-based combinations.

[0129] In the post-processing of each processing path **102**, of the embodiment of FIG. **3**, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by majority formation, sum formation or also by multiplication of successive probability values.

[0130] Further, by clustering multiple results, for example, it is possible to detect which objects of multiple detected objects lying close to one another are the same object so that they are not incorrectly detected multiple times.

[0131] Likewise, it is possible to multiply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic sub-results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic sub-results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic sub-results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.

[0132] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Here, weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.

[0133] For example, the result of an object classification in which the object type **117** of an object disposed in the wall **105** is defined can be utilized to increase the weight of a wall type classification in which the wall type **123** of the respective wall **105** is determined in the post-processing at locations without objects **113**, because the respective radar data **103** at these locations are less influenced by reflections of the objects **113**.

[0134] FIG. 5 shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0135] Graphs a) and b) of FIG. 5 show two different alternatives of a joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.

[0136] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network.

[0137] The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107**, based on this.

[0138] In graph b), in contrast, the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.

[0139] In an additional concatenation layer **148**, the partial results of the partial diagnoses of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.

[0140] The correspondingly configured diagnostic module **107** is designed to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.

[0141] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known from the prior art.

[0142] FIG. 6 shows a further schematic illustration of a measurement of the measuring device **100** according to one embodiment.

[0143] FIG. 6 shows radar data **103** captured by the measuring device **100** as the measuring device **100** moves along the direction of movement **153** relative to the wall **105** along a movement path

179. The radar data **103** is plotted in FIG. **6** for different frequencies **181**. Further, a window **167** is shown in FIG. **6**. The window **167** shows a contiguous spatial area extending over a sub-area of the movement path **179** and comprising radar data **103** and radar signals of a plurality of different frequencies **181**.

[0144] Only one window **167** is represented in FIG. **6**, by way of example. As the measuring device **100** moves along the movement path **169**, corresponding windows **167** are recorded at predetermined distances, i.e., at predetermined measurement points. The windows **167** may be disposed as overlapping, wherein the overlap region representing spatial areas of the movement path **169** through at least two windows **167**.

[0145] Also shown in FIG. **6** is a measurement signal **182**. The measurement signal **182** may be based, for example, on an object **113** disposed in the wall **105**.

[0146] To examine the wall, based on the received radar data **103** of the radar sensor unit **101** of the measurement device **100**, which respectively maps the wall **105** to be diagnosed and if applicable the objects **113** disposed in the wall **105**, the diagnostic module **107** performs the wall diagnostics and generates the diagnostic results **109**.

[0147] In the wall diagnostics, the wall type classification is performed and the wall type **123** of the wall **105** is determined by the diagnostic module **107** based on the radar data.

[0148] Further, the object recognition of the object **113** disposed in the wall **105** is performed based on the radar data **103** and taking into account the wall type classification results, i.e. the determined wall type **123**. The object recognition includes at least the object detection, i.e., determining the object position **115**, and the object classification, i.e., determining the object type **117** of the object **113**.

[0149] Further, the diagnostic results **109** of the display unit **111** of the measuring device **100** are provided and displayed therein. The diagnostic results comprise at least the determined wall type **123** and, if an object **113** is present, the object position **115** and the object type **117**.

[0150] The analysis of the radar data **103** is carried out in the windows **167** shown above. For each window, a partial classification of the radar data **103** is performed and partial classification results of the wall type **123** are determined. The results of the wall type **123** determined for the different windows **167** are subsequently merged to an overall result in a merger.

[0151] The partial classification results represent results for different sub-areas of the wall **105**. Through the merger, the partial classification results may be summarized into the overall result. The overall result is representative of a contiguous area of the wall generated by summarizing the sub-areas of the partial classification results.

[0152] The merger can be carried out in this case taking into account a weighting. The weighting may take wall type classification results into consideration with a lower weighting, such as those that deviate greatly from the wall types **123** determined for windows recorded at earlier times **167**. This can prevent erroneous results from adversely affecting the wall diagnostics.

[0153] The determined wall type classification results of the wall type **123** determined for a window **167** recorded at the current time may then be compared to stored wall types **123** determined for windows **167** that were recorded at earlier times. If the comparison shows that the wall type **123** determined for a predetermined number of windows **167** deviates from that determined for the windows **167** recorded at earlier times, then the wall type **123** determined for the windows **167** recorded at later times is stored as the current wall type **123** and used for the wall diagnostics.

[0154] If a deviating wall type **123** is determined for current windows **167**, but if the number of windows **167** for which the deviating wall type **123** was determined is less than the predetermined number, then the deviating wall type **123** is not considered and the wall diagnostics are continued using the wall type **123** determined based on the windows **167** which were determined earlier.

[0155] Further, movement data of the movement detection unit **141** is included when the measuring device **100** is moved along the movement path **179** and considered for determining a position of the

measuring device **100** relative to the wall **105**. The position of the measuring device **100** determined in this way relative to the wall **105** is considered in the wall diagnostics, in particular in the object recognition, in order to determine at least the object position **115** of the object **113** disposed in the wall **105**.

[0156] When movement of the measuring device **100** relative to the wall **105** is detected, the wall diagnostics are performed by the diagnostic module **107** only at predefined measurement points. The measurement points are thereby defined by predefined distances that the measuring device **100** has traversed between the measurement points, i.e., between the performances of the wall diagnostics by the diagnostic module **107** relative to the wall **105**, along the movement path **179**.

[0157] If the measuring device **100** has not yet reached the next measurement point during the movement along the movement path **179**, i.e. the measuring device **100** has not yet traveled the predefined distance, the radar data **103** recorded up to that point will be stored. The stored radar data **103** is then processed by the diagnostic module **107** only upon reaching the next measurement point.

[0158] FIG. 7 shows a flowchart of a method **200** for operating a measuring device **100** according to one embodiment.

[0159] To operate the measuring device **100**, in a first method step **201**, radar data **103** of the radar sensor unit **101** of the measuring device **100** is first received, wherein the radar data **103** depicts the wall **105** to be diagnosed.

[0160] In a further method step **203**, the wall diagnostics are performed by performing the analysis of the radar data **103** generating diagnostic results **109** using the diagnostic module **107** of the measuring device **100**.

[0161] For this purpose, in a method step **205** the wall type classification is performed and the wall type **123** of the wall **105** is determined.

[0162] For this purpose, in a method step **211**, a partial classification is performed on individual windows **167** of the radar data **103** and partial classification results for the wall type **123** are provided.

[0163] In a method step **213**, the partial classification results are merged to an overall classification result and the wall type **123** is determined based on the overall classification result.

[0164] In a further method step **217**, the wall type classification results of the wall type **123** are stored.

[0165] In a method step **219**, the wall type classification results of the wall type **123** determined at a current time are compared with the wall type classification results for the wall type **123** generated and stored at earlier times.

[0166] In a method step **221**, the wall type classification results for the wall type **123** determined at the current time are provided, if the current wall type classification results for the wall type **123** deviate from the stored wall type classification results for the wall type **123** of a predetermined number of windows **167** of the radar data **103**.

[0167] In a further method step **239**, otherwise the stored wall type classification results for the wall type **123** are provided.

[0168] In a further method step **207**, the object recognition is performed by the diagnostic module **107** based on the radar data **103** and taking into account the wall type classification results of the wall type **123**. The object recognition comprises at least the object detection and the object classification.

[0169] In another method step **209**, the diagnostic results **109** of the diagnostic module **107** are provided to a display unit **111** for displaying the diagnostic results **109**. The diagnostic results **109** comprise at least the object position **115** and the object type **117** of the object **113** disposed in the wall and the wall type **123** of the wall **105** to be examined.

[0170] FIG. 8 shows a further flowchart of the method **200** for operating a measuring device **100** according to a further embodiment.

[0171] The embodiment shown is based on the embodiment in FIG. 7 and comprises all the method steps described there.

[0172] In deviation from the embodiment in FIG. 7, in a method step **223**, the movement data of the movement detection unit **141** of the measuring device **100** is first received. The movement data here depicts the movement of the measuring device **100** relative to the wall **105** between a plurality of positions of the measuring device **100** relative to the wall **105**.

[0173] In a method step **225**, a check is carried out based on the movement data to determine whether the movement of the measuring device **100** relative to the wall **105** comprises a predefined distance.

[0174] If no movement by the predefined distance of the measuring device **100** relative to the wall **105** has been detected, the recorded radar data **103** is stored in a method step **227** and a new check is performed.

[0175] Conversely, if a corresponding movement of the measuring device **100** relative to the wall **105** by the predefined distance is detected, then the wall diagnostics is performed in method step **203**.

[0176] Here, in a method step **231**, the object depth determination is performed by the diagnostic module **107** and the object depth **119** of the object **113** in the wall **105** is determined.

[0177] Alternatively or additionally, in a method step **233**, the object extension determination is performed by the diagnostic module **107** and the object extension **121** of the object **113** is determined.

[0178] Further, in a method step **235**, the radar data **103** is pre-processed by the first pre-processing module **135** and input data **133** is provided to the wall type classification module **129**.

[0179] Further, in a further method step **237**, pre-processing of the radar data **103** is performed by the second pre-processing module **137**, taking into account the provided wall type **123**. Further, input data **133** is provided to the object recognition module **131** by the second pre-processing module.

[0180] FIG. 9 shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method **200** for operating a measuring device **100**.

[0181] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can in this case be any desired storage medium known from the prior art.

Claims

1. A computer-implemented method of operating a measuring device, comprising: receiving radar data of a radar sensor unit of the measuring device, wherein the radar data depicts a wall to be diagnosed; performing a wall diagnostics by performing an analysis of the radar data and generating diagnostic results using a diagnostic module of the measuring device, wherein the performing the wall diagnostics comprises: performing a wall type classification and providing wall type classification results of a wall type of the wall using the diagnostic module based on the radar data; performing an object recognition of an object disposed in the wall using the diagnostic module based on the radar data and taking into account the wall type classification results of the wall type, wherein the object recognition comprises an object detection and an object classification; and providing the diagnostic results of the diagnostic module to a display unit of the measuring device for displaying the diagnostic results to a user of the measuring device, wherein the diagnostic results comprise at least one object position of the object in the wall and an object type of the object and/or the wall type of the wall.

2. The method according to claim 1, wherein the received radar data comprises a plurality of radar signals of different frequencies reflected by the wall, and wherein the received radar data is

summarized to a plurality of windows.

- 3.** The method according to claim 2, wherein performing the wall type classification comprises: performing partial classifications on individual windows of the radar data and providing partial classification results of the wall type; and merging a plurality of partial classification results to an overall classification result and determining the wall type based on the overall classification result.
 - 4.** The method according to claim 3, wherein the merger is effected by performing a merger algorithm, and/or wherein, during the merger, the partial classification results are considered with a weighting with respect to the results of the object recognition.
 - 5.** The method according to claim 1, wherein performing the wall type classification comprises: storing the wall type classification results for the wall type; comparing the currently determined wall type classification results for the wall type with stored wall type classification results for the wall type; and providing the current wall type classification results for the wall type if the current wall type classification results for the wall type deviate from the stored wall type classification results for the wall type for a predetermined number of windows of the radar data.
 - 6.** The method according to claim 1, wherein: the radar data is received for a plurality of positions of the measuring device relative to the wall, and the method further comprises receiving movement data of a movement detection unit of the measuring device, wherein the movement data represents a movement of the measuring device relative to the wall between the plurality of positions of the measuring device relative to the wall.
 - 7.** The method according to claim 6, further comprising: checking, based on the movement data of the movement detection unit, whether a movement of the measuring device relative to the wall by a predefined distance has been detected; if no movement has been detected, storing the radar data and performing the check again; and if movement by the predefined distance has been detected, performing the wall diagnostics based on the radar data using the diagnostic module.
 - 8.** The method according to claim 1, wherein the wall diagnostics further comprises: summarizing sub-objects detected and classified at various positions of the measuring device relative to the wall, based on radar data to a contiguous object, if there is a predefined distance dependency between the sub-objects.
 - 9.** A computing unit configured to perform the method of operating a measuring device according to claim 1.
 - 10.** A computer program product comprising instructions which, when the program is executed by a data processing unit, cause the data processing unit to perform the steps of the method for operating a measuring device according to claim 1.
 - 11.** The method according to claim 1, wherein the measuring device includes a wall diagnostic device.
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